

Assignment – 14

1. Install the `redux` and `react-redux` packages in your existing React project and set up a basic Redux store in a new file called `store.js`.
2. Create a Redux action called `addSong` that takes a song name as payload and a reducer called `playlistReducer` that adds the song to an array in the store.
3. Build a simple React component called `PlaylistManager` that lets users type a song name into an input and dispatches the `addSong` action to update the Redux store, displaying the current playlist below.
4. Add a `removeSong` action and update your `playlistReducer` so users can remove a song from the playlist by clicking a 'Remove' button next to each song.

Hint: Use the song's index or unique name as an identifier when dispatching `removeSong`.

Answer:-

`Store.js`

```
import { createStore } from "redux";
import playlistReducer from "../reducers/playlistReducer";

const store = createStore(playlistReducer);

export default store;
```

`PlaylistAction.js`

```
export const addSong = (songName) => {
  return {
    type: "ADD_SONG",
    payload: songName,
  };
};

export const removeSong = (index) => {
  return {
    type: "REMOVE_SONG",
    payload: index,
  };
};
```

PlaylistReducer.js

```
const initialState = {
  playlist: [],
};

const playlistReducer = (state = initialState, action) => {
  switch (action.type) {
    case "ADD_SONG":
      return {
        ...state,
        playlist: [...state.playlist, action.payload],
      };

    case "REMOVE_SONG":
      return {
        ...state,
        playlist: state.playlist.filter(
          (_, index) => index !== action.payload
        ),
      };

    default:
      return state;
  }
};

export default playlistReducer;
```

main.jsx

```
import React from "react";
import ReactDOM from "react-dom/client";
import App from "./App";

import { Provider } from "react-redux";
import store from "./store";

ReactDOM.createRoot(document.getElementById("root")).render(
  <Provider store={store}>
    <App />
  </Provider>
);
```

PlaylistManager.jsx

```
import { useState } from "react";
import { useDispatch, useSelector } from "react-redux";

import {
  addSong,
  removeSong,
} from "../actions/playlistActions";

const PlaylistManager = () => {
  const [songName, setSongName] = useState("");

  const dispatch = useDispatch();

  const playlist = useSelector(
    (state) => state.playlist
  );

  const handleAddSong = () => {
    if (!songName.trim()) return;

    dispatch(addSong(songName));

    setSongName("");
  };

  const handleRemoveSong = (index) => {
    dispatch(removeSong(index));
  };

  return (
    <div style={{ padding: "20px" }}>
      <h1>Playlist Manager</h1>

      <input
        type="text"
        placeholder="Enter song name"
        value={songName}
        onChange={(e) =>
          setSongName(e.target.value)
        }
      />

      <button
        onClick={handleAddSong}
        style={{ marginLeft: "10px" }}
      >
        Add Song
      </button>
    </div>
  );
};
```

```

    <hr />

    <h2>Current Playlist</h2>

    {playlist.length === 0 ? (
      <p>No Songs Added</p>
    ) : (
      <ul>
        {playlist.map((song, index) => (
          <li key={index}>
            {song}

            <button
              onClick={() =>
                handleRemoveSong(index)
              }
              style={{
                marginLeft: "10px",
              }}
            >
              Remove
            </button>
          </li>
        ))}
      </ul>
    )}
  </div>
);
};

export default PlaylistManager;

```

App.jsx

```

import PlaylistManager from "../components/PlaylistManager";

function App() {
  return (
    <div>
      <PlaylistManager />
    </div>
  );
}

export default App;

```